

Needs Expression

AI-Based Intrusion Detection & Prevention System Integrating Port Polymorphism (Aegis-Morph) & Honeypot Mechanisms

Module : Artificial Intelligence | Status : Academic Project

Project Team

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Project Contextualisation

■ 1. General Context

The digital transformation of organizations across all sectors — public administration, healthcare, finance, industry, and education — has led to an unprecedented expansion of networked infrastructures. Whether deployed as Local Area Networks (LAN), Wide Area Networks (WAN), Metropolitan Area Networks (MAN), wireless networks, cloud environments, or hybrid architectures, these infrastructures now serve as the backbone of critical operations, enabling resource sharing, real-time communication, and distributed service delivery.

This ubiquity, however, comes at a cost: the larger and more complex a network, the broader its attack surface. Network security has consequently become one of the most pressing challenges in modern information systems management. Malicious actors continuously evolve their strategies, exploiting vulnerabilities at every layer of the network stack — from the physical and data-link layers up through the application layer.

In response, the field of **Network Intrusion Detection and Prevention Systems (IDS/IPS)** has emerged as a critical discipline. Traditional approaches rely on predefined signatures and static rule sets. While effective against known threats, these systems fail to anticipate novel, adaptive, or previously unseen attack patterns. The limitations of such approaches have created a clear and growing demand for **intelligent, self-learning security systems** capable of analyzing network behavior dynamically and responding autonomously.

■ 2. The Role of Artificial Intelligence in Network Security

Artificial Intelligence, and more specifically machine learning, offers a fundamentally different approach to intrusion detection. Rather than relying on manually curated rules, AI-based systems learn statistical

models of normal network behavior from labeled traffic datasets. Any significant deviation from the learned baseline is flagged as a potential threat — enabling the detection of both known and previously uncharacterized attack vectors.

This paradigm shift brings several decisive advantages over classical IDS/IPS solutions: adaptability to evolving threats, the ability to process and classify high-volume traffic in real time, reduced dependency on manual signature updates, and the capacity to support autonomous decision-making — including automated threat containment without human intervention.

Classical IDS/IPS	AI-Based IDS/IPS
Relies on predefined signatures	Learns behavioral patterns from data
Fails against unknown/novel attacks	Detects anomalies beyond known signatures
Requires constant manual updates	Self-adapts through model retraining
No autonomous response capability	Supports automated threat neutralization
High false positive rate at scale	Optimized precision/recall via ML tuning
Static and network-topology-specific	Generalizable across diverse network types

■ 3. Project Framework & Methodology

This project is carried out within the **Artificial Intelligence** academic module by a team of five students. Its objective is to design, implement, test, and validate an AI-powered IDS/IPS system, deployed over a simulated network topology spanning multiple nodes and traffic types. The system must be capable of monitoring network traffic in real time, detecting suspicious behavioral patterns, profiling threat actors, triggering alerts, and enforcing automated countermeasures.

The project is structured around the **PDCA (Plan–Do–Check–Act)** continuous improvement methodology, which ensures that each development phase is rigorously planned, executed, evaluated, and refined before progression to the next stage.

Plan	Define project objectives, identify data sources, design the network topology, select AI models and evaluation criteria, and structure the development roadmap.
Do	Implement the full system pipeline: data preprocessing, model training, real-time traffic capture, threat detection engine, dashboard, and automated response module.
Check	Evaluate the system against simulated threat scenarios. Measure performance against predefined metrics (accuracy, latency, false positive rate). Document observations and gaps.
Act	Validate findings through cross-validation and end-to-end testing. Identify limitations, propose improvements, and consolidate results into the final academic deliverable.

■ 4. Design Methodology : CPS (Context – Problematic – Solution)

Beyond the overarching PDCA framework, the project adopts the **CPS (Context–Problematic–Solution)** scientific design methodology to ensure that every technical decision

is grounded in observed, field-validated needs rather than assumptions.

Context	Establishes the environment in which the project operates: the nature of modern network infrastructures, the evolving threat landscape, and the limitations of existing security tools. This Needs Expression document constitutes the Context phase.
Problematic	Derived from structured interviews conducted with network and infrastructure professionals. An interview guide will be developed, administered, and transcribed into formal interview reports. The technical problematic — including the specific threats to address — will be formalized exclusively from the insights gathered during this phase.
Solution	A targeted technical solution is proposed based on the findings of the Problematic phase. It specifies the chosen approach, tools, datasets, model architecture, evaluation metrics, and the integration of advanced defense mechanisms into the IDS/IPS system.

■ 5. Network Scope

The system is designed to be applicable across a broad spectrum of network architectures and deployment contexts. While the academic prototype will be validated in a controlled simulation environment, the principles, models, and mechanisms developed are intended to generalize to real-world infrastructure of any type and scale.

Network Type	Description	Relevance
LAN	Local Area Network — controlled, high-speed, limited geographic scope	Primary simulation environment for prototype validation
WAN	Wide Area Network — geographically distributed, multi-site connectivity	Relevant for enterprise and inter-organizational threat detection
MAN	Metropolitan Area Network — city-scale infrastructure	Applicable to smart city, utility, and campus-wide deployments
Wireless / Wi-Fi	IEEE 802.11 networks — mobile, access-point-based	Relevant to BYOD environments and open-access networks
Cloud / Virtual Networks	Software-defined and cloud-hosted network infrastructure	Growing target surface for network-level intrusions
Hybrid Architectures	Combination of on-premise and cloud, wired and wireless	Reflects most real-world organizational deployments

■ 6. Advanced Security Concepts Integrated in the Project

Beyond the core AI-based detection engine, this project integrates two advanced proactive defense mechanisms that significantly enhance the depth and sophistication of the overall security posture:

■ Port Polymorphism via Aegis-Morph

Aegis-Morph is a dynamic port reassignment mechanism that continuously rotates the listening ports of protected network services according to a defined schedule or trigger condition. By making service port assignments unpredictable and transient, this technique disrupts an attacker's ability to map the network and identify exploitable entry points through conventional probing strategies. The AI detection engine is

designed with full awareness of the current port assignment state, ensuring that legitimate traffic to dynamically assigned ports is not misclassified as suspicious.

■ Honeypot Integration

A honeypot is a deliberately exposed decoy system deployed within the network infrastructure. It mimics the appearance of a legitimate host or service while containing no actual operational data or functionality. Any interaction with the honeypot is, by definition, unauthorized and constitutes a high-confidence threat signal. Honeypot-captured events enrich the AI engine's threat intelligence, provide ground-truth labeled data for attacker behavioral analysis, and enable the generation of comprehensive attacker profiles including intent, methodology, and originating context.

■ 7. Expected System Capabilities

Capability	Description
Real-Time Traffic Monitoring	Continuous capture and analysis of network flows with live dashboard indicators
AI-Driven Threat Detection	Machine learning model trained on labeled network traffic datasets to identify abnormal behaviors
Attacker Profiling	Automatic generation of a threat actor profile (network identity, behavior pattern, threat level)
Graduated Alert System	Multi-level alerting mechanism triggered proportionally to the confidence and severity of detected threats
Automated Blocking	Autonomous enforcement of firewall countermeasures upon confirmed threat detection, without human intervention
Port Polymorphism (Aegis-Morph)	Dynamic rotation of service port assignments to increase the cost and complexity of network reconnaissance
Honeypot-Based Deception	Decoy node that captures unauthorized interaction attempts and feeds behavioral data to the AI engine
Cross-Network Applicability	Architecture designed for generalization across LAN, WAN, wireless, cloud, and hybrid environments